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Exploring the Digital Transformation of Bangladesh's Education System: A Critical Narrative Review of ICT Integration

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Authors' contributions

This work was carried out in collaboration between both authors. Both authors read and approved the final manuscript.

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Abstract

Bangladesh has pursued information and communication technology in education through digital content, multimedia classrooms, teacher-development initiatives, online platforms and, more recently, an explicit policy commitment to blended learning. Yet the system's digital transformation remains uneven, with substantial differences between policy ambition, institutional capacity and learners' actual opportunities to benefit. This critical narrative review examines the trajectory of technology integration across primary, secondary and higher education, evaluates the implementation barriers revealed before and during the coronavirus disease 2019 disruption, and develops priorities for a more equitable and educationally consequential transformation. Literature published principally from 2009 to 21 May 2026 was identified through live searches of accessible scholarly indexes, journal and DOI records, institutional repositories and authoritative national and international sources, supplemented by citation searching. The evidence indicates that Bangladesh has expanded the visibility and acceptability of digital education, but much of the empirical literature measures attitudes, intentions and self-reported experiences rather than learning outcomes,

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institutional performance or long-term equity effects. The strongest recurring constraints are unreliable connectivity and electricity, device and data affordability, multidimensional digital exclusion, weak pedagogical integration, uneven teacher support, limited institutional readiness, fragile maintenance arrangements and misalignment between digital delivery, curriculum and assessment. Pandemic-era online learning accelerated experimentation but also exposed a system dependent on household resources and urban connectivity. Studies of teacher development suggest that school-embedded, iterative and context-sensitive support is more credible than short technical training, while higher-education studies consistently favour blended over fully online provision. Future progress therefore depends less on adding isolated platforms than on building a resilient learning ecosystem that guarantees minimum access, strengthens teacher and leadership capability, supports Bangla and accessible content, governs data and artificial intelligence responsibly, and evaluates whether technology improves learning, inclusion and system responsiveness.

Keywords: *Blended learning; digital divide; educational technology; ICT in education; teacher professional development; Bangladesh; education policy; digital governance.*

1. Introduction

Digital transformation in education is frequently reduced to the acquisition of devices, conversion of textbooks into digital formats or movement of lectures onto online platforms. These activities may be necessary, but they do not in themselves constitute transformation. A transformed education system changes how learning is designed, how teachers are supported, how institutions use information, how assessment is organised, and how learners participate across differences of geography, gender, disability and socio-economic position. International evidence also cautions against assuming that technology has an intrinsically positive educational effect. Meta-analytic and experimental syntheses find that technology can support learning, but the average effect is conditional on pedagogy, implementation quality, instructional time, teacher mediation and the counterfactual against which it is compared (Escueta et al., 2020; Tamim et al., 2011). This distinction is especially important for lower-middle-income systems, where digital initiatives operate within constraints that are not external to educational design but constitutive of it. Recent Bangladesh-specific design research similarly indicates that meaningful ICT integration depends on context-sensitive instructional planning that links available resources, curriculum and learner participation rather than on hardware provision alone (Hwang & Lim, 2024).

Bangladesh offers a consequential case. During the past decade and a half, national policy has linked education modernisation with the wider Digital Bangladesh agenda, promoted multimedia classrooms and digital content, expanded technology-supported teacher development, and encouraged higher education institutions to adopt learning management systems and online delivery. The National Education Policy framed information and communication technology (ICT) as an essential educational capability, while later planning documents positioned digital platforms, content repositories, institutional information systems and blended approaches as components of sector development (Ministry of Education, 2010, 2019). In higher education, the University Grants Commission's blended-learning policy formalised a post-pandemic direction in which face-to-face and digitally mediated learning are intended to operate as an integrated system rather than as emergency substitutes (University Grants Commission of Bangladesh, 2022). These developments demonstrate policy continuity, but they do not establish that implementation has been coherent or that benefits have been evenly distributed. Recent work on Bangladesh Open University further frames learning-management-system-integrated blended learning as an institutional design task requiring coordinated pedagogical, technical and strategic planning (Ahmed et al., 2025).

The coronavirus disease 2019 (COVID-19) disruption became an unplanned stress test of this policy trajectory. Educational closure affected tens of millions of learners, yet participation in remote learning was limited and sharply differentiated by place and level of education. A nationally reported survey found that only 18.7% of school-age children participated in remote learning during school closures, with lower participation in rural than urban areas and substantial variation across levels of schooling (Bangladesh Bureau of Statistics & UNICEF, 2023). The emergency therefore accelerated institutional experimentation while exposing the dependence of digital education on household devices, affordable data, electricity, study space and adult support. University students reported access problems, weak preparedness, difficulties maintaining attention and reduced interaction, while qualitative studies described assessment uncertainty, insufficient training and psychological

strain (Al-Amin et al., 2021; Bashir et al., 2021; Saha et al., 2021). The crisis did not create the digital divide, but it made the divide educationally visible.

Bangladesh-specific scholarship has grown rapidly since 2020. Much of it documents university students' perceptions, technology acceptance, online-class experiences and institutional responses. This evidence is useful because it captures a period of abrupt change, yet it is methodologically concentrated in cross-sectional surveys, convenience samples and urban or university settings. Studies frequently model behavioural intention among respondents who already possess sufficient connectivity to complete online questionnaires. Such designs can explain variation among connected users but tend to under-represent those most excluded from digital learning. In addition, positive intention or perceived usefulness is often interpreted as evidence of successful transformation even though acceptance is not equivalent to sustained use, effective pedagogy or improved learning. Technology Acceptance Model and Unified Theory of Acceptance and Use of Technology constructs are valuable for studying adoption, but their explanatory reach narrows when institutional governance, affordability, social inequality and instructional quality are treated as peripheral variables (Davis, 1989; Venkatesh et al., 2003).

The literature outside higher education is thinner but analytically important. Earlier work on technology-enhanced teacher development in rural and non-elite settings suggests that digital resources can support pedagogical change when they are embedded in school practice, accompanied by collegial support and adapted to local conditions (Jamil, 2018; Shohel & Banks, 2012). Secondary-level studies, particularly in English language teaching and Bangladesh and Global Studies, show generally favourable teacher attitudes but persistent obstacles in infrastructure, training, content alignment and confidence (Khan & Kuddus, 2020; Mallick & Salam, 2024). These findings complicate a simple deficit narrative. Teachers are not necessarily resistant to technology; rather, motivation often coexists with weak technical support, limited preparation time and pedagogical models that do not match classroom realities. The central implementation problem is therefore not merely whether teachers and students accept technology, but whether the system creates the organisational conditions in which meaningful use is feasible. Recent secondary-school evidence likewise shows that teacher readiness, curriculum guidance and access to appropriate digital resources remain interdependent conditions for classroom ICT integration (Farhana et al., 2024).

Existing reviews and conceptual papers often focus on online learning during COVID-19, higher education e-learning, or broad descriptions of digital opportunities and constraints. They rarely integrate the full policy-to-practice chain across educational levels, distinguish emergency remote teaching from planned digital transformation, or appraise what kinds of evidence underpin claims of progress. A system-level synthesis must connect infrastructure, teacher capability, curriculum, assessment, institutional leadership, household resources, data governance and political economy. It must also confront the unevenness of the evidence base itself: higher education is over-represented, while primary education, madrasas, technical and vocational education and training (TVET), learners with disabilities, low-connectivity regions and long-term learning outcomes receive less rigorous attention. A recent study of Bangladeshi polytechnic students also extends the evidence base to TVET, a subsystem that remains comparatively under-represented in national digital-education research (Sadam & Al Mamun, 2024).

1.1 Scope, Research Gap and Objectives

This review examines the digital transformation of Bangladesh's education system with emphasis on ICT integration in teaching, learning, teacher development, institutional delivery and educational governance. It covers primary, secondary and higher education, while drawing attention to the comparatively limited evidence on madrasas, TVET, open and non-formal education, disability inclusion and educational administration. The review distinguishes planned digital and blended learning from emergency remote teaching, and it treats access as a multidimensional condition involving devices, connectivity, affordability, skills, support and beneficial outcomes. Its objectives are to reconstruct the policy and implementation trajectory, critically assess evidence on integration and user experience, explain persistent policy–practice gaps, evaluate emerging opportunities and governance risks, and propose research and policy priorities that are proportionate to the strength of available evidence. The distinctive contribution is an integrated system perspective that links adoption research with pedagogical quality, equity, institutional capacity and long-term resilience rather than treating digitalisation as a sequence of isolated technology projects.

2. Methods for Literature Selection

2.1 Review Design and Rationale

A critical narrative review was selected because the research question concerns a heterogeneous policy and implementation field rather than a single intervention with comparable outcomes. The evidence includes qualitative studies, cross-sectional surveys, technology-acceptance models, mixed-methods institutional studies, policy documents, national surveys and conceptual frameworks. A meta-analysis would have been inappropriate because definitions, populations, technologies and outcomes differ substantially, and a systematic effectiveness review would have excluded much of the contextual and governance evidence required to explain implementation. Narrative synthesis nevertheless requires explicit searching, appraisal and evidence calibration. The review was guided by the principles of relevance, transparency, critical engagement and balanced interpretation reflected in the Scale for the Assessment of Narrative Review Articles (SANRA) (Baethge et al., 2019).

2.2 Information Sources

Live searches were conducted through accessible scholarly web search, the Directory of Open Access Journals, ERIC records, PubMed and PubMed Central where relevant, journal and institutional repositories, DOI landing pages and authoritative national and international websites. Bibliographic identity was checked against journal pages, DOI records or reputable repositories. National policy and statistical sources were obtained from the Ministry of Education, University Grants Commission of Bangladesh, Bangladesh Bureau of Statistics and UNICEF. International contextual sources were obtained from UNESCO and the World Bank. Backward citation searching from relevant reviews and empirical papers, forward citation searching through accessible scholarly records and related-article searching were used to identify earlier and newer work. No claim is made that Scopus, Web of Science or other subscription databases were comprehensively searched.

2.3 Search Strategy and Search Period

The principal search period was January 2009 to 21 May 2026, chosen to capture the expansion of Digital Bangladesh-era education initiatives, the maturation of mobile and platform-based learning, the COVID-19 disruption and the subsequent blended-learning policy phase. Foundational studies published before 2009 were retained when they supplied established theoretical or evaluative frameworks. Representative Boolean combinations included (“Bangladesh” AND education AND (“digital transformation” OR ICT OR “educational technology” OR e-learning OR “online learning” OR “blended learning”)); (“Bangladesh” AND teacher* AND (ICT integration OR “digital competence” OR “technology-enhanced professional development”)); (“Bangladesh” AND student* AND (“digital divide” OR access OR affordability OR readiness OR “technology acceptance”)); and (“Bangladesh” AND education AND (policy OR governance OR implementation OR assessment OR equity OR disability OR rural)). Separate searches targeted primary, secondary, higher, madrasa and technical education, as well as learning management systems, mobile learning, digital content, cybersecurity, data protection and artificial intelligence.

2.4 Eligibility Criteria

Eligible academic sources were peer-reviewed journal articles and methodologically relevant scholarly reviews that directly addressed education technology in Bangladesh or provided an established framework needed for interpretation. Authoritative policy, statistical and sector reports were included when they provided national context not available from journal studies. English-language publications were prioritised because reliable translation and metadata verification were necessary. Studies were considered across educational levels and methods, but inclusion required enough bibliographic information to verify title, authorship, outlet, date and DOI or stable official location. Sources were excluded when they were promotional, unverifiable, duplicated, based only on commentary, or too weakly related to support the claim for which they might be cited. Preprints were not used where peer-reviewed evidence was available.

2.5 Screening, Duplicate Handling and Study Selection

Titles and abstracts were first assessed for substantive relevance to ICT integration, online or blended learning, digital inequality, teacher capability, institutional readiness, educational governance or technology-related

outcomes in Bangladesh. Full texts or sufficiently detailed publisher abstracts were then examined to determine study population, design, analytical contribution and claim relevance. Duplicate records were reconciled using DOI, title, author and year. When repositories and journal pages reported different online and issue years, the final published issue metadata was used. Inaccessible records were retained only when authoritative metadata and a sufficiently detailed abstract could be verified; sources with unresolved bibliographic ambiguity were removed. No screening totals are reported because the search was iterative and thematic rather than a closed systematic-review protocol.

2.6 Critical Appraisal and Evidence Synthesis

Evidence was appraised according to design appropriateness, sampling and representativeness, construct validity, transparency, attention to confounding, correspondence between data and conclusions, and transferability across educational levels and regions. Particular caution was applied to online surveys likely to exclude disconnected students, cross-sectional acceptance models unable to establish sustained behaviour, self-reported learning outcomes, and studies conducted during emergency conditions but interpreted as evidence about normal online education. Qualitative studies were valued for explaining institutional and household mechanisms, although small purposive samples limited prevalence claims. Themes were developed iteratively around policy architecture, subsystem implementation, digital inequality, teacher and institutional capability, pedagogical quality, governance and future resilience. Contradictory findings were interpreted in relation to differences in population, modality, timing, infrastructure and measurement rather than averaged into a false consensus.

3. Conceptualising Digital Transformation in Education

Digital transformation is best understood as a socio-technical and institutional process. Digitisation converts analogue resources into digital form; digitalisation uses digital tools to modify processes; transformation redesigns relationships among pedagogy, organisation, data and participation. Bangladesh has advanced in the first two domains through digital content, multimedia classrooms, online admission and administration, video-based teacher support and learning platforms. The evidence that these changes have consistently altered classroom interaction, assessment or learning outcomes is less secure. This distinction matters because the visible presence of technology may coexist with conventional teacher-centred pedagogy. A projector displaying a digital textbook can reproduce the instructional logic of a chalkboard, while a learning management system can become a repository rather than an environment for feedback, collaboration and adaptive support.

Several analytical traditions help explain this gap. Technology-acceptance models foreground perceived usefulness, ease of use, social influence and facilitating conditions (Davis, 1989; Venkatesh et al., 2003). They are well suited to questions about willingness to adopt platforms and have been widely applied in Bangladeshi higher education (Anam et al., 2026; Maisha & Shetu, 2023; Miah et al., 2023; Tamal et al., 2022). Yet acceptance models tend to locate the dependent variable within individual intention even when constraints arise from institutional financing, curriculum, labour conditions or network infrastructure. A student may value online learning but remain unable to afford data; a teacher may intend to use a platform but lack time, technical support or assessment authority. Adoption is therefore a necessary but insufficient indicator of transformation.

Barrier frameworks offer a complementary lens. Ertmer's distinction between first-order barriers, such as access, time, equipment and technical support, and second-order barriers, such as beliefs, confidence and pedagogical orientation, remains useful because it prevents infrastructure and teacher agency from being collapsed into one explanation (Ertmer, 1999). Hew and Brush (2007) broadened this analysis to include institutional culture, subject culture, assessment and leadership. Bangladesh evidence suggests that the two barrier orders interact rather than occur sequentially. Repeated connectivity failures can reduce confidence and reinforce lecture-based use; limited pedagogical models can make expensive infrastructure educationally unproductive; and weak maintenance can turn an initially successful intervention into a short-lived project.

Digital-divide theory also needs to move beyond a binary distinction between connected and unconnected users. Van Dijk (2006) conceptualised access as a sequence involving motivation, material access, skills and usage, while Helsper (2012) linked digital exclusion to broader social fields and offline disadvantage. Applied to Bangladesh, this implies that a household with one smartphone may be counted as connected even when the device is shared, the data package is limited, the screen is unsuitable for extended study and the learner lacks a

quiet space. The educational divide emerges through unequal capacity to convert nominal connectivity into sustained, supported and consequential learning. This interpretation explains why broad mobile penetration did not translate into universal remote-learning participation.

Blended-learning theory provides a more credible strategic direction than an unqualified shift to online education. Blended learning is not the simple coexistence of face-to-face and online components but the intentional integration of modalities to exploit their different strengths (Garrison & Kanuka, 2004). Institutional adoption requires alignment among strategy, structure, support, professional development and evaluation (Graham et al., 2013). Bangladesh's policy turn towards blended learning is therefore promising, but only if institutions move from emergency scheduling to instructional redesign. Table 1 summarises the conceptual distinctions used throughout this review.

Table 1. Conceptual distinctions for analysing digital transformation in Bangladesh's education system

Concept	Operational meaning	Common misinterpretation	Implication for evaluation	Supporting references
Digitisation	Conversion of content or records into digital formats.	Equating digital files with pedagogical change.	Assess accessibility, usability, updating and integration into teaching.	Graham et al. (2013); UNESCO (2023)
Digitalisation	Use of digital tools to modify teaching, administration or communication processes.	Counting devices, platforms or log-ins as evidence of effectiveness.	Examine process quality, reliability, workload and participation.	Escueta et al. (2020); Tamim et al. (2011)
Digital transformation	System-level redesign of pedagogy, organisation, data use, assessment and participation.	Treating a portfolio of separate projects as a coherent system.	Evaluate institutional capability, learning, equity and sustainability together.	Garrison and Kanuka (2004); Graham et al. (2013)
Technology acceptance	Perceived usefulness, ease, social influence and intention to use.	Assuming intention demonstrates learning effectiveness or inclusion.	Connect adoption measures to sustained use and educational outcomes.	Davis (1989); Venkatesh et al. (2003)
Multidimensional digital divide	Inequality in motivation, devices, connectivity, skills, support, usage and outcomes.	Classifying any mobile or internet access as adequate educational access.	Use disaggregated measures of meaningful access and benefit.	Helsper (2012); van Dijk (2006)
Emergency remote teaching	Temporary continuity response under crisis conditions.	Interpreting crisis practices as mature online education.	Separate emergency constraints from planned programme quality.	Bond et al. (2021); Rapanta et al. (2020)

Note. ICT = information and communication technology. The distinctions are analytical rather than mutually exclusive; a single initiative may operate at more than one level

4. Policy Architecture and Implementation Trajectory in Bangladesh

Bangladesh's digital-education trajectory has been shaped by a broader state-modernisation agenda. The National Education Policy 2010 treated ICT literacy and use as educational priorities and anticipated the development of digital teaching materials, teacher capability and institutional facilities (Ministry of Education, 2010). Subsequent planning translated this direction into programmes involving multimedia classrooms, digital content, teacher portals, connectivity and education information systems. The 2019 progress review of the Master Plan for ICT in Education in Bangladesh (2012–2021) assessed a sector-wide architecture extending from infrastructure and content to teacher capacity, management and implementation coordination (Ministry of Education, 2019). These policies are important because they moved ICT from an optional enrichment activity

towards a system expectation. They also created a persistent implementation challenge: objectives expanded faster than the institutional capacity to coordinate procurement, maintenance, professional learning, monitoring and instructional redesign.

The policy architecture has often been project-centred. Projects can demonstrate feasibility, mobilise resources and produce visible outputs, but they also risk fragmentation. A multimedia classroom, teacher portal, online course platform and administrative database may be financed and governed separately even though their educational value depends on interoperability and common support structures. When initiatives are evaluated primarily through deployment indicators, policy can overestimate transformation. The relevant question is not only how many institutions received equipment or how many teachers attended training, but whether teachers use the resources regularly, whether content aligns with curriculum, whether learners participate, and whether the arrangement can be maintained after project funding or external technical assistance ends.

COVID-19 disrupted this gradual trajectory. Institutions were compelled to adopt online classes, television lessons, social media communication and improvised assessment with limited preparation. Higher education moved more rapidly because universities had greater autonomy, more adult learners and comparatively stronger connectivity, but public-private and urban-rural differences remained. Studies from this period show both adaptive capacity and systemic weakness. Private universities and some public institutions established synchronous classes and learning management systems, while many students faced device sharing, unstable networks, high data costs and unsuitable home environments (Al-Amin et al., 2021; Alom et al., 2023; Bashir et al., 2021; Shohel et al., 2022). The pandemic therefore functioned as an accelerator of adoption and a diagnostic of unequal readiness.

It is analytically necessary to distinguish continuity from quality. Emergency provision was valuable when the alternative was prolonged disconnection, but rapid replication of classroom lectures through video conferencing did not automatically deliver effective online pedagogy. International research on emergency remote teaching emphasises teacher presence, purposeful learning activities, interaction and support rather than transmission alone (Bond et al., 2021; Rapanta et al., 2020). Bangladeshi evidence similarly identifies limited interaction, assessment difficulty, reduced attention and weak feedback as recurring concerns. These problems cannot be attributed solely to teachers; they arose from compressed preparation, limited institutional guidance, uncertainty over regulations and the design constraints of low-bandwidth participation.

The post-pandemic blended-learning policy represents an attempt to convert improvised practices into an institutional model. The University Grants Commission policy provides legitimacy for combining physical and online modes and encourages institutions to develop infrastructure, content, staff capability and quality assurance (University Grants Commission of Bangladesh, 2022). Its significance lies less in declaring a new modality than in requiring universities to make enduring arrangements. Implementation will depend on whether institutions establish minimum standards, workload recognition, technical and instructional-design support, accessible content, valid assessment and data governance. Without these elements, blended learning may become a label attached to irregular online supplementation.

Policy coherence also depends on the relationship between emergency experience and long-term reform. Comparative evidence from Bangladesh and Nepal shows that institutions adopted online education through different combinations of preparedness, improvised practice and local support, making national labels poor indicators of actual capability (Shrestha et al., 2022). Arefin et al. (2023) similarly argued for a sustainable framework that connects emergency protocols with infrastructure, pedagogy and governance. These contributions are valuable because they shift attention from isolated classroom problems to system design, but their broad frameworks still require operational specification. Responsibilities must be allocated among ministries, directorates, regulators and institutions; standards need financing and enforcement mechanisms; and monitoring must identify whether weaker institutions are converging or falling further behind. A transformation policy that defines national direction without clarifying these implementation relationships may increase formal compliance while preserving practical fragmentation.

Table 2 shows that the trajectory has progressed from policy recognition and resource provision towards emergency expansion and formalised blending. The phases are cumulative rather than fully sequential: basic infrastructure deficits persist even as artificial intelligence (AI), analytics and cybersecurity enter policy

discussion. Bangladesh therefore faces a compressed transformation in which unresolved first-generation problems coexist with emerging governance risks.

Table 2. Major phases in Bangladesh's digital-education policy and implementation trajectory

Phase	Dominant emphasis	Principal achievements	Persistent weakness	Supporting references
Policy recognition and foundational provision (2010s)	ICT literacy, digital content, multimedia classrooms and teacher exposure.	ICT became a stated component of national education modernisation.	Deployment was often weakly connected to pedagogy, maintenance and evaluation.	Ministry of Education (2010); Shohel and Banks (2012)
Scaling and system planning (late 2010s)	Long-term master planning, platforms, connectivity, information systems and professional capacity.	A broader system architecture was articulated.	Coordination, recurrent financing and institution-level capability remained uneven.	Jamil (2018); Ministry of Education (2019); Sarker et al. (2019)
Emergency remote teaching (2020–2021)	Continuity through video conferencing, television, messaging and online materials.	Rapid experimentation and increased acceptance of digital delivery.	Severe participation inequality, limited readiness and pedagogical improvisation.	Al-Amin et al. (2021); Badiuzzaman et al. (2021); Bangladesh Bureau of Statistics and UNICEF (2023)
Formal blended-learning turn (from 2022)	Institutional integration of physical and online learning.	A policy basis for sustained blended provision in higher education.	Implementation standards, workload, accessibility and quality assurance are still developing.	University Grants Commission of Bangladesh (2022); Shohel et al. (2022)
Data- and AI-enabled transformation (emerging)	Analytics, generative AI, automation, cybersecurity and digital services.	Potential for personalisation, decision support and wider access.	Privacy, bias, academic integrity, procurement and capacity risks may outpace governance.	UNESCO (2023); Harun Or Rashid and Mustafa (2026)

Note. The dates indicate dominant tendencies, not sharp transitions. Earlier infrastructure and capability problems continue in later phases

5. ICT Integration across Educational Subsystems

Digital transformation is not proceeding uniformly across Bangladesh's educational subsystems. Differences in governance, learner age, teacher employment, institutional autonomy, assessment and household support shape what technologies can be used and how. The evidence base itself is uneven: higher education accounts for a disproportionate share of peer-reviewed studies, while primary and secondary evidence more often emerges from programme evaluations, teacher studies and institutional reports. Consequently, confidence about national patterns is greater for access barriers than for comparative educational effectiveness.

At primary level, digital resources have been used for teacher professional development, classroom content and remote continuity. The English in Action-related literature is particularly instructive because it treats technology as a support for school-based professional learning rather than as a replacement for teachers. Shohel and Banks (2012) found that portable digital resources embedded in school contexts could support changes in classroom practice, while Jamil (2018) showed that implementation depended on layered contextual factors, including stakeholder relationships, local resources and institutional conditions. These studies provide a stronger theory of change than programmes based on one-off training: teachers require repeated opportunities to observe, practise,

discuss and adapt pedagogy. Yet the evidence is based on specific interventions and cannot be assumed to represent all primary schools.

The pandemic revealed the limits of household-dependent remote education for younger learners. Television and low-tech communication can achieve greater reach than synchronous online classes, but participation depends on programme awareness, scheduling, adult mediation and the fit between content and learners' routines. National survey evidence indicates that remote-learning engagement was low overall and lower in rural settings (Bangladesh Bureau of Statistics & UNICEF, 2023). This finding challenges strategies that define provision by broadcast availability rather than verified participation. For primary education, transformation must therefore include school-community communication, offline resources, local support and mechanisms for re-engagement, not only digital delivery.

At secondary level, the available studies show a recurring divergence between favourable attitudes and constrained practice. English-language teachers have recognised the potential of ICT for authentic materials, audiovisual input and learner engagement but have reported inadequate equipment, unreliable internet, limited training and curriculum pressures (Khan & Kuddus, 2020; Tasneem, 2022). Teachers of Bangladesh and Global Studies similarly expressed positive attitudes while identifying access and institutional barriers (Mallick & Salam, 2024). Imon (2024) found that teachers' perceptions of classroom digitalisation were shaped by capability and local conditions. These studies are predominantly perception-based, so they establish willingness and perceived constraints more strongly than they establish effects on attainment.

Secondary education also faces high-stakes assessment and large-class constraints. Digital activities that require extended device access, individual feedback or open-ended inquiry may be difficult to reconcile with examination-oriented curricula and crowded classrooms. Technology can support visualisation, practice and formative assessment, but it may also reinforce passive presentation if teacher performance is judged mainly by syllabus coverage. A credible strategy must align digital pedagogy with assessment reform and provide subject-specific models. Generic ICT competence is insufficient because productive technology use in science, languages, social studies and vocational subjects demands different representations, tasks and feedback.

Higher education has experienced the most visible expansion. Before the pandemic, studies described e-learning and virtual classrooms as promising but emergent, with infrastructure and organisational readiness limiting adoption (Chowdhury, 2020b; Sarker et al., 2019). Chowdhury (2020a) advocated flipped and blended approaches but also identified prerequisites such as institutional planning and teacher readiness. During the pandemic, institutions rapidly expanded online classes, exposing variation in preparedness, participation and classroom activity (Al-Amin et al., 2021). Qualitative evidence from public-university English departments identified access, logistics, pedagogy, curriculum, assessment and psychological challenges (Bashir et al., 2021). Later studies found continuing interest in digital and blended learning but persistent concern about interaction, infrastructure and quality (Al Masud et al., 2023; Rouf et al., 2024).

Technology-acceptance studies provide a consistent but incomplete picture. Perceived usefulness, ease of use, facilitating conditions, social influence and attitude commonly predict intention to use online platforms (Anam et al., 2026; Maisha & Shetu, 2023; Miah et al., 2023; Tamal et al., 2022). These findings support investment in usability, support and confidence. Their limitation is that samples are often drawn from private universities or students already participating online, and outcomes are intentions rather than verified academic performance. Models can therefore inform adoption strategy but should not be used as substitutes for programme evaluation.

Institutional differentiation within higher education requires closer attention. Public universities, private universities, National University-affiliated colleges and specialised institutions operate under different financing, governance and student-residence conditions. A platform that works in a well-resourced private university with small classes and dedicated technical staff may be impractical in a large affiliated college. Conversely, public institutions may possess strong disciplinary expertise but face procurement delays, crowded classrooms and limited support staffing. Existing studies tend to aggregate these differences or sample a small group of institutions, limiting generalisability. Future implementation should therefore use tiered standards: a universal baseline for access, accessibility, security and quality, followed by institution-specific pathways that reflect programme type and capability. Such differentiation is not an argument for lower expectations; it is a means of preventing uniform mandates from producing inequitable compliance burdens.

The evidence for madrasas, TVET, open education, non-formal learning and learners with disabilities is comparatively sparse. This absence is substantively important. TVET requires practical demonstration, workplace simulation and equipment access; madrasa systems vary in governance and curriculum; open and non-formal providers may have extensive distance-education experience; and disability inclusion requires captioning, screen-reader compatibility, accessible assessment and assistive technologies. A system review cannot infer that findings from universities generalise to these settings. Their under-representation is itself a research and policy gap.

Across subsystems, participation is shaped by age and dependence. University students can often manage passwords, asynchronous deadlines and platform navigation independently, whereas younger learners need adult mediation and regular teacher contact. Adolescent girls may have less control over shared devices or study time, while learners with disabilities may encounter inaccessible interfaces even when connectivity is adequate. Rural learners can face both network constraints and reduced access to technical assistance. These mechanisms imply that universal platform provision can produce unequal educational exposure. Implementation plans should specify who supports the learner, how inaccessible content is reported, what alternative format is available and how absence is followed up. Without these operational safeguards, digital provision risks becoming an institutional record of delivery rather than evidence of learning opportunity.

Administrative digitalisation has received less critical educational scrutiny than teaching platforms. Student information systems, online admissions, examination processing, teacher portals and education management information systems can improve coordination and transparency, but benefits depend on data quality, interoperability and routine use in decisions. Poorly integrated systems increase reporting burden and create parallel databases. Administrative transformation should therefore be judged by whether it reduces transaction costs, supports timely intervention and strengthens accountability without compromising privacy. Table 3 synthesises the evidence by subsystem.

Table 3. ICT integration across educational subsystems: evidence, confidence and unresolved issues

Subsystem	Most documented uses	What the evidence supports	Principal evidence limitation	Supporting references
Primary education	Digital content, multimedia lessons, teacher development and remote continuity.	Context-embedded technology can support professional learning; household-dependent remote participation is fragile.	Few long-term, nationally representative studies of classroom use and learning outcomes.	Jamil (2018); Rahman and Siddik (2022); Shohel and Banks (2012)
Secondary education	Multimedia presentation, language resources, online communication and teacher ICT use.	Teacher attitudes are often positive, but infrastructure, training and curriculum pressures restrict use.	Perception studies predominate; subject-specific outcome evidence is limited.	Imon (2024); Khan and Kuddus (2020); Mallick and Salam (2024); Tasneem (2022)
Higher education	Synchronous classes, learning management systems, virtual classrooms, mobile learning and blended courses.	Adoption expanded rapidly; users value flexibility but report interaction, readiness and access problems.	Urban, private-university and connected-student samples are over-represented.	Al-Amin et al. (2021); Al Masud et al. (2023); Rouf et al. (2024); Sarker et al. (2019)
Madrasa, TVET and non-formal education	Digital content and remote support are reported, but rigorous evidence is limited.	Potential is plausible but subsystem-specific effectiveness cannot be inferred from university evidence.	Major gap in peer-reviewed implementation and outcome research.	Ministry of Education (2019); UNESCO (2023)
Educational administration	Admissions, records, examinations, portals and management information systems.	Digital processes can improve timeliness and coordination when data are interoperable and used.	Educational value, workload, data quality and privacy effects are rarely evaluated together.	Ministry of Education (2019); Bhatta et al. (2019)

Note. TVET = technical and vocational education and training

6. Persistent Implementation Challenges

The dominant implementation challenges are often presented as separate deficits, yet they form a mutually reinforcing system. Connectivity affects pedagogy; affordability affects participation; teacher capability affects perceived usefulness; leadership affects support; assessment affects instructional design; and household conditions affect attention and persistence. Treating each problem through a separate procurement or training activity produces partial solutions that can fail at their interfaces.

Infrastructure remains the most visible constraint. Studies across levels report unstable internet connections, electricity interruption, limited device access and weak technical support (Al-Amin et al., 2021; Badiuzzaman et al., 2021; Islam & Habib, 2022). These problems are not uniform. Urban institutions may have campus connectivity while students return to rural homes with weak networks; a family may own a smartphone but have no computer; and a device may be shared among siblings. Bandwidth requirements also matter. A policy that assumes continuous video participation imposes a higher cost and reliability threshold than one designed around downloadable materials, audio, text and asynchronous interaction. Infrastructure policy should therefore specify educational service levels rather than count connections alone.

Affordability is a distinct issue from physical availability. Mobile data costs, device replacement, charging, maintenance and software subscriptions shift recurrent expenses to households and institutions. During emergency online learning, students who could technically connect still reported difficulty sustaining participation because of data cost and device limitations (Badiuzzaman et al., 2021; Das, 2021). The latent digital divide is consequently dynamic: a learner may participate intermittently, avoid cameras, miss assessment or rely on borrowed equipment. Binary access measures underestimate this precarity. Subsidies or zero-rated educational resources may reduce costs, but they can also create restricted internet environments if learners are confined to selected platforms. Policy should balance affordability with open access and portability of learning materials.

Digital inequality mirrors and can intensify existing inequality. Rural residence, household income, gendered control of devices, disability, language and institutional type influence the capacity to benefit. National evidence that remote-learning participation was much lower than universal availability claims demonstrates the difference between supply and uptake (Bangladesh Bureau of Statistics & UNICEF, 2023). University surveys likely understate the problem because the most disconnected students are less likely to respond. Saha et al. (2021) also linked digital exclusion and emergency online learning to psychological distress among Dhaka undergraduates, showing that inequality is experienced not only as missed content but as anxiety, uncertainty and social disconnection. Equity assessment should therefore include participation, continuity, learning, wellbeing and opportunity cost.

Teacher capability is frequently described as digital literacy, but this term can obscure the distinction between operating tools and designing learning. A teacher may be competent with presentation software yet lack models for online discussion, formative feedback, collaborative inquiry or accessible assessment. Bangladesh evidence indicates that teachers need subject-specific, practice-based and continuing support (Jamil, 2018; Rahman & Siddik, 2022; Shohel & Banks, 2012). Short workshops can increase familiarity but rarely alter pedagogy without coaching, peer learning, leadership support and time to redesign materials. Training should also be differentiated: novice users require basic confidence, while experienced users need instructional design, data literacy, accessibility and ethical AI capability.

Professional learning should be organised as an implementation system rather than a sequence of courses. Teachers need access to exemplars that demonstrate how technology serves a specific learning objective, opportunities to rehearse and receive feedback, and communities in which problems can be solved collectively. School leaders require parallel development so that timetables, observation and resource allocation reinforce the expected practice. Digital platforms can extend coaching and peer exchange, but participation should not become an additional unpaid burden. Recognition through workload, certification or career progression may improve continuity, although incentives must not reward superficial log-in activity. The appropriate performance evidence is the quality of tasks, feedback and learner participation, not the number of digital tools

used. This approach also allows training to evolve as platforms change without tying professional competence to a particular vendor.

Institutional leadership determines whether individual capability becomes routine practice. Universities that moved online rapidly often depended on motivated teachers, informal peer support and commercially available platforms. Such improvisation is adaptive but fragile. Sustainable provision requires help desks, instructional designers, accessible-content guidance, learning analytics expertise, cybersecurity, procurement capacity and clear responsibility for quality. School leaders likewise need to schedule shared resources, protect teacher preparation time and create a culture in which experimentation is supported rather than punished. The implementation literature consistently shows that facilitating conditions influence adoption, but facilitating conditions should be understood as organisational arrangements, not merely device availability (Anam et al., 2026; Miah et al., 2023; Venkatesh et al., 2003).

Curriculum and assessment can either enable or block digital pedagogy. Emergency online classes often transferred lectures without redesigning tasks or feedback. Students then experienced reduced engagement and difficulty understanding lessons (Al-Amin et al., 2021; Rahman et al., 2023). Online assessment raised concerns about integrity, connectivity and validity, prompting reliance on attendance, assignments or oral examinations. A transformation strategy should avoid treating surveillance technology as the primary answer. Assessment design can reduce misconduct through authentic tasks, staged submissions, oral defence, varied question banks and formative feedback, while preserving accessibility and trust. In schools, alignment with high-stakes examinations is essential; teachers will not sustain inquiry-oriented digital activities if examinations reward recall alone.

Quality assurance must consequently evaluate the learning design rather than the mere existence of an online component. Course review should examine whether outcomes, activities and assessment are aligned; whether students receive timely feedback; whether materials meet accessibility and bandwidth requirements; and whether participation expectations are realistic. Institutions also need procedures for students to challenge technical failure, inaccessible content or unfair online assessment. National quality frameworks can set common principles, but peer review at programme level is needed to interpret them. The quality burden should be proportionate: elaborate documentation can overwhelm institutions with limited staff, while weak oversight allows digital provision to become inconsistent. A staged maturity model linked to support and improvement would be more useful than a binary approval process.

Content quality and language are additional constraints. Bangladesh has developed substantial digital resources, but quantity does not ensure curricular coherence, accessibility or classroom usability. Content should be available in Bangla and, where appropriate, minority languages; optimised for low bandwidth; compatible with assistive technologies; and structured for teacher adaptation. Central repositories can support scale, while local creation can improve relevance. The governance challenge is to combine quality assurance with teacher agency. Overly centralised content can become static and disconnected from learners, while uncurated user-generated materials can reproduce error and inequity.

Student engagement depends on social and instructional presence, not only connectivity. Studies report distraction, reduced attention, limited interaction and difficulty maintaining routines (Alom et al., 2023; Karim et al., 2025; Rouf et al., 2024). These findings are sometimes interpreted as evidence that online learning is inherently inferior. A more careful interpretation is that prolonged synchronous lecture, weak feedback and home constraints produce disengagement. Well-designed asynchronous tasks, small-group interaction and timely feedback can improve participation, but they require teacher workload and institutional support. Blended learning is likely to be more acceptable because it restores social contact while retaining flexibility, but poorly designed blending can simply duplicate workload.

Maintenance and recurrent financing receive less attention than initial deployment. Equipment failure, obsolete software, password management, licence renewal and turnover of trained staff can erode implementation after launch. Sustainable budgeting must include replacement cycles, connectivity, support staff, accessibility testing and content updating. Donor- or project-funded pilots should be evaluated for total cost of ownership and transition to recurrent budgets. Table 4 presents the interacting mechanisms through which these barriers affect educational outcomes.

Table 4. Interacting implementation barriers and their educational mechanisms

Barrier domain	Mechanism of educational disadvantage	Why common responses are insufficient	More defensible system response	Supporting references
Connectivity, electricity and devices	Interrupted participation, reduced interaction, missed assessment and inability to access rich media.	Device distribution alone does not ensure data, power, maintenance or shared-use availability.	Define minimum service levels, provide low-bandwidth and offline options, and budget recurrent support.	Al-Amin et al. (2021); Badiuzzaman et al. (2021)
Affordability and household conditions	Intermittent participation, device sharing, unsuitable study space and opportunity costs.	Nominal household internet access masks precarious or unequal educational use.	Measure meaningful access and target subsidies, community access and downloadable resources.	Bangladesh Bureau of Statistics and UNICEF (2023); van Dijk (2006)
Teacher capability and beliefs	Technology reproduces lecture transmission or is avoided under pressure.	One-off technical workshops do not produce subject-specific pedagogical change.	Use school-embedded coaching, peer communities, exemplars and protected design time.	Ertmer (1999); Jamil (2018); Shohel and Banks (2012)
Institutional leadership and support	Individual innovation remains inconsistent and fragile.	Policy mandates without staffing, workload and technical support shift risk to teachers.	Create instructional-design, help-desk, accessibility and cybersecurity functions.	Graham et al. (2013); Sarker et al. (2019)
Curriculum and assessment	Digital activities become peripheral when examinations reward different learning.	Platform adoption does not resolve validity, feedback or integrity problems.	Align curriculum, formative assessment and authentic assessment with modality.	Bashir et al. (2021); Rapanta et al. (2020)
Equity, language and disability	Existing social disadvantages are translated into unequal participation and benefit.	Uniform platforms assume homogeneous devices, language and sensory access.	Apply universal design, Bangla/local-language resources and disaggregated monitoring.	Helsper (2012); UNESCO (2023)
Maintenance and recurrent finance	Equipment and platforms deteriorate after initial deployment.	Capital-focused projects understate total cost of ownership.	Fund replacement, licences, support, security and content renewal within recurrent budgets.	Bhatta et al. (2019); Ministry of Education (2019)

Note. The domains are mutually reinforcing; interventions should be evaluated as packages rather than isolated inputs

7. Explaining the Policy–Practice Gap

Bangladesh's policy–practice gap is not adequately explained by a single shortage. Policy documents articulate broad ambitions, while implementation occurs through institutions with different resources, incentives and authority. The gap emerges because technology projects often define success through inputs and adoption, whereas educational systems produce outcomes through repeated interaction among curriculum, teachers, learners, leadership, households and assessment.

First, visible inputs are politically and administratively easier to count than changes in teaching. Devices, classrooms, connections and training events generate clear completion indicators. Pedagogical improvement is slower, context-dependent and difficult to attribute. This creates an implementation bias towards procurement and platform launch. The bias is not unique to Bangladesh, but resource constraints magnify its consequences because recurrent support is easily crowded out by capital expenditure. Tamim et al. (2011) and Escueta et al. (2020) show that the educational effects of technology vary across implementation models; average positive effects do not justify assuming that any technology investment will improve learning.

Second, projects are frequently designed at a level above the institution but rely on local adaptation. Central coordination is needed for standards, purchasing and equitable coverage, yet schools and universities require discretion to match tools to subject, learners and infrastructure. When central policy is highly prescriptive, local actors may comply symbolically. When policy is too general, well-resourced institutions advance while weaker institutions stagnate. The appropriate design is bounded autonomy: national minimum standards and shared services combined with institution-level implementation plans, support and evaluation.

Bounded autonomy requires a reciprocal accountability arrangement. Central agencies should publish interoperable standards, provide shared infrastructure where scale is advantageous and target additional support to low-capacity institutions. Local institutions should document how they adapt the standards, report participation and accessibility problems, and use evidence to improve provision. Monitoring should be diagnostic rather than punitive because institutions may otherwise conceal failures or inflate usage. Independent evaluation and public reporting can strengthen accountability, but indicators must avoid league tables that reward affluent institutions for advantages outside their control. The aim is to make variation visible, direct support towards bottlenecks and create learning across institutions. This governance design would also reduce the recurring problem of pilots that cannot be compared because each uses different definitions of access, participation and success.

Third, training is often separated from organisational change. Teachers may attend workshops but return to classrooms without reliable equipment, peer support or recognition of preparation time. The earlier Bangladesh teacher-development studies demonstrate why context matters: technology-supported learning was productive when resources were integrated with school-based practice and professional relationships (Jamil, 2018; Shohel & Banks, 2012). The implication is not merely that more training is needed, but that professional learning must be embedded in work. Universities similarly require instructional-design support and workload models that recognise the labour of producing accessible digital courses.

Fourth, policy has sometimes conflated adoption with educational value. During and after the pandemic, studies commonly measured intention to use, satisfaction or perceived usefulness. Such measures are legitimate, but they sit early in the causal chain. Positive attitudes do not show that students learned more, persisted longer or experienced greater inclusion. Conversely, low satisfaction during crisis delivery does not prove that planned blended learning is ineffective. A mature evaluation framework should connect inputs, implementation fidelity, participation, teaching quality, short-term learning, progression, equity and cost. Few Bangladesh studies follow learners longitudinally or compare alternative delivery models with robust controls.

Fifth, the evidence ecosystem is itself skewed. Online surveys are efficient and were often the only feasible method during school closures, but they select for connectivity and literacy. Private-university studies may capture advanced institutional practice while under-representing public colleges and resource-constrained regions. Qualitative studies reveal mechanisms but cannot establish national prevalence. Policy synthesis must therefore avoid turning a large number of similar cross-sectional studies into an illusion of strong causal evidence.

Sixth, transformation is affected by political economy and market structure. Institutions increasingly depend on commercial platforms, cloud services and telecommunications. These services can accelerate implementation, but contracts may create vendor lock-in, recurrent foreign-currency costs, limited data portability and uncertain

privacy obligations. Open-source platforms can reduce licensing dependence but still require skilled administration. Procurement should therefore assess interoperability, accessibility, data location, exit options, security and total cost, not only initial price or feature lists.

Finally, education technology competes with other urgent sector needs. Bangladesh's education expenditure and institutional capacity remain constrained, so digital investment must demonstrate opportunity cost and distributional value (Bhatta et al., 2019). A high-cost platform used mainly by already advantaged students may have lower system value than basic connectivity, teacher support or remedial learning. Digital transformation should be treated as a means of improving educational capability rather than as a parallel modernisation agenda insulated from learning priorities.

8. Emerging Opportunities and Governance Risks

The limitations of current implementation do not imply that digital transformation should be slowed indiscriminately. They indicate that the next phase should be more selective, evidence-driven and governance-aware. Bangladesh has several advantages: widespread familiarity with mobile devices, a large youth population, experience with national digital services, expanding institutional use of platforms and a policy basis for blended learning. These assets can support a resilient model if technology is designed around actual constraints.

A mobile-first but not mobile-only strategy is particularly relevant. Smartphones can extend communication, microlearning, formative quizzes, audio-visual explanation and access to repositories. Noor (2024) found continuing interest in mobile learning among Bangladeshi university students, while broader pandemic evidence shows the practical importance of phones. Yet mobile-only provision can constrain writing, coding, data analysis and extended reading. Institutions should design responsive, low-bandwidth materials while maintaining access to larger-screen devices through campuses, libraries and community hubs. Downloadable packages and synchronisation can reduce dependence on continuous connectivity.

Blended learning offers the most plausible mainstream model. It can reserve face-to-face time for discussion, laboratory work, mentoring and social connection while using digital environments for preparation, practice, feedback and continuity. Bangladesh studies generally show stronger support for a combined model than for permanently replacing classrooms (Al Masud et al., 2023; Chowdhury, 2020a; Tahura et al., 2025). The risk is superficial blending, in which teachers deliver the same content twice or add online tasks without workload adjustment. Course redesign should begin with learning outcomes and decide which modality best supports each activity.

Open educational resources and local content could improve affordability and relevance. Shared repositories reduce duplication and enable teachers to adapt high-quality materials. Their value depends on licensing, version control, curriculum tagging, accessibility and mechanisms for peer review. Bangladesh can build on existing digital-content experience by shifting from static resources towards modular learning objects, teacher guides, diagnostic tasks and reusable assessment. Translation into Bangla and support for local languages should be treated as educational infrastructure, not an optional localisation stage.

Data use can strengthen early warning, resource allocation and feedback. Learning management systems and administrative platforms generate information about participation, submission and progression. These data can help institutions identify disengagement, but log-in frequency is an imperfect proxy for learning. Analytics should be validated, interpreted by educators and combined with non-digital information. Automated risk scores may reproduce bias when students with unstable connectivity appear less engaged. Governance must specify purpose limitation, access control, retention, correction and human review.

Generative AI creates both educational opportunities and risks. It can support explanation, translation, formative feedback, lesson planning and accessibility, especially where teacher workloads are high. At the same time, it can generate inaccurate content, reproduce linguistic and social bias, expose personal data and destabilise assessment. Bangladesh should avoid both uncritical adoption and blanket prohibition. Institutions need guidance on acceptable use, disclosure, verification, intellectual property, academic integrity and age-appropriate safeguards. Teacher education should include AI literacy as critical evaluation rather than prompt operation alone.

Data governance should precede large-scale AI integration rather than follow it. Schools and universities need inventories of what learner and staff data are collected, which vendors process them, where they are stored, how

long they are retained and how errors can be corrected. Consent is not an adequate sole basis in compulsory education because learners may have little practical choice. Procurement should require data minimisation, security testing, deletion and export provisions, and restrictions on secondary model training. Algorithmic outputs used for grading, discipline, admission or risk prediction should remain contestable and subject to human review. These safeguards are not barriers to innovation; they are conditions for maintaining trust and preventing the costs of experimentation from falling disproportionately on students with less institutional power.

Cybersecurity becomes more important as institutions integrate identity, assessment, finance and learning data. A 2026 study of zero-trust readiness in Bangladeshi higher education illustrates growing attention to institutional security, but readiness frameworks require adaptation to budgets and staffing (Harun Or Rashid & Mustafa, 2026). Security controls that are too complex can exclude users or encourage unsafe workarounds; weak controls can expose student records and disrupt services. Shared national or sectoral services may help smaller institutions access expertise, incident response and secure identity management.

A resilient education ecosystem should combine high-tech and low-tech channels. Future crises may involve floods, cyclones, health emergencies or political disruption. Continuity planning should therefore include downloadable content, television and radio, messaging, printed materials, community access points and mechanisms for teachers to maintain contact. The aim is not to preserve identical instruction under every condition but to maintain a minimum learning relationship and enable rapid recovery. Table 5 translates these opportunities into implementation prerequisites, risks and monitoring indicators.

Table 5. Strategic directions for equitable and governed digital transformation

Strategic direction	Implementation prerequisite	Principal risk	Priority monitoring question	Supporting references
Minimum viable digital learning guarantee	Defined standards for device access, affordable connectivity, offline materials, support and accessibility.	A nominal guarantee may be met without meaningful learner use.	Can every learner access and complete core activities under realistic household conditions?	Bangladesh Bureau of Statistics and UNICEF (2023); Badiuzzaman et al. (2021)
Pedagogy-led blended learning	Course redesign, teacher support, workload recognition and aligned assessment.	Blending may duplicate lectures and increase workload without improving learning.	Which activities improve participation, feedback and learning in each modality?	Chowdhury (2020a); Garrison and Kanuka (2004); Graham et al. (2013)
Mobile-first, multi-device access	Low-bandwidth design, downloadable resources and campus/community access to larger devices.	Mobile-only design can narrow complex academic tasks.	Are students able to perform the full range of required learning activities?	Noor (2024); Al-Amin et al. (2021)
Open, local-language and accessible resources	Open licensing, curriculum tagging, peer review, Bangla/local-language production and universal design.	Repositories can become large but unused, outdated or inaccessible.	Are materials used, adapted, accessible and associated with better teaching?	UNESCO (2023); Khan and Kuddus (2020)
Responsible learning analytics and AI	Purpose limitation, data minimisation, human oversight, verification and AI literacy.	Bias, privacy loss, inaccurate feedback and assessment distortion.	Do automated systems improve decisions without disproportionately harming vulnerable learners?	UNESCO (2023); Harun Or Rashid and Mustafa (2026)
Resilient multi-channel continuity	Pre-agreed low-tech and high-tech channels, contact protocols and recovery plans.	Emergency channels may reach households without producing participation.	Can institutions maintain contact, formative support and re-entry during disruption?	Bond et al. (2021); Rahman and Siddik (2022)

Note. AI = artificial intelligence. Monitoring questions should be disaggregated by gender, disability, location, income proxy and institution type where ethically and legally appropriate

9. Future Directions

The immediate research priority is to move from intention and perception studies towards learning and equity outcomes. Bangladesh needs longitudinal and quasi-experimental evaluations that compare well-specified digital or blended models with credible alternatives. Studies should report implementation fidelity, teacher support, learner exposure, attrition and subgroup effects rather than only average satisfaction. Where randomisation is feasible, it should be used for discrete interventions such as adaptive practice, tutoring or teacher-coaching models. Where it is not, phased roll-outs, matched comparisons and interrupted time-series designs can strengthen inference. Outcome sets should include attainment, retention, feedback quality, attendance, wellbeing and cost, with follow-up long enough to detect whether initial engagement persists.

A second priority is an equity-disaggregated national monitoring framework. Current administrative indicators should be extended from infrastructure ownership towards meaningful educational access. Measures need to capture whether learners have an appropriate device when needed, sufficient data or offline access, accessible content, support, and the ability to complete required tasks. Data should be disaggregated by location, gender, disability, socio-economic proxy, language, institution type and level while protecting privacy. Repeated measurement would reveal whether gaps narrow or are merely hidden by aggregate connectivity growth.

Teacher professional development requires comparative implementation research. The most credible existing evidence favours school-embedded and context-sensitive support, but the relative contribution of coaching, peer communities, digital exemplars, leadership and incentives remains uncertain. Cluster-randomised or stepped-wedge designs could compare feasible support packages across districts, while qualitative process evaluation explains why effects vary. Teacher outcomes should extend beyond self-efficacy to observed instructional practice, task quality, feedback and learner participation. Similar research is needed for university faculty, where instructional design and assessment capability are central.

Under-represented subsystems should become a deliberate research focus. Primary and secondary research needs nationally varied samples and direct classroom observation. Madrasas and TVET require modality-specific studies, particularly for practical, laboratory and workshop learning. Open and non-formal education can provide lessons on distance support and flexible pacing. Disability-inclusive research should involve learners and organisations of persons with disabilities in co-design, test compatibility with assistive technologies and report accessibility failures as implementation outcomes rather than exceptional cases.

Cost and sustainability analysis should accompany effectiveness research. Studies should estimate total cost of ownership, including connectivity, support staff, maintenance, replacement, licences, content renewal, accessibility and cybersecurity. Cost-effectiveness comparisons can identify when low-tech options, shared infrastructure or teacher support offer greater value than sophisticated platforms. Procurement pilots should include exit and interoperability tests so that scaling decisions account for data portability and vendor dependence.

The emerging AI agenda requires prospective governance research. Bangladesh institutions need evidence on how students and teachers use generative AI in Bangla and English, where errors and bias occur, and how AI changes assessment behaviour. Controlled studies can evaluate AI-supported feedback or tutoring, but deployment should include human oversight, adverse-event reporting and privacy review. Policy research should compare institutional guidance, identify feasible disclosure standards and test assessment designs that preserve valid demonstration of learning.

A national research infrastructure would make these priorities more feasible. Common definitions, open instruments and secure data-linkage protocols could reduce duplication and improve comparability across projects. Universities, teacher-education institutions and government agencies could maintain a registry of major education-technology evaluations, including null and negative findings, to limit publication bias. Research partnerships should involve practitioners and learners in problem definition and interpretation, while preserving independence from vendors and implementing agencies. Where administrative data are used, access should follow transparent ethical and privacy procedures. A shared evidence platform would allow policy to distinguish context-specific adaptation from failure, identify scalable components and update standards as technology changes. Such an infrastructure would also strengthen local methodological capacity rather than relying repeatedly on externally designed pilots.

A final priority is implementation science at system scale. Research should examine how national policy is interpreted by directorates, universities, districts, school leaders and teachers; which bottlenecks recur; and how adaptation affects outcomes. Mixed-methods programme theory, realist evaluation and comparative case studies are appropriate because the same technology may succeed under one configuration of support and fail under another. The strategic objective is not to identify a universally superior platform, but to learn which combinations of infrastructure, pedagogy, leadership, financing and governance produce durable improvement in specific contexts.

10. Conclusions

Bangladesh has established a substantial policy and institutional foundation for digital education, and the COVID-19 period accelerated awareness, experimentation and acceptance. The most defensible conclusion is nevertheless that transformation remains partial. Progress is clearest in the expansion of digital content, online communication, institutional platforms and policy recognition of blended learning. Evidence is weaker for consistent improvements in learning, inclusion, assessment quality and system efficiency. Much of the scholarship captures perceptions and intentions among connected university populations, so confidence about nationwide educational effects must remain cautious.

The principal obstacles are systemic rather than purely technological. Connectivity, electricity, affordability, device sharing, teacher capability, institutional support, curriculum, assessment, maintenance and household conditions interact. Isolated responses therefore produce limited gains. Teacher-development evidence shows that technology becomes educationally useful when it is embedded in practice, supported over time and adapted to context. Higher-education evidence indicates that students and teachers value flexibility but generally require the interaction and structure of blended provision. Equity cannot be inferred from mobile penetration or platform availability; it must be demonstrated through sustained participation and beneficial outcomes across social groups.

The next phase should shift from project accumulation to a governed learning ecosystem. A minimum digital-learning guarantee, pedagogy-led blended design, low-bandwidth and offline options, accessible Bangla and local-language resources, institutional support functions, interoperable data systems and responsible AI governance are mutually dependent components. Investment decisions should be judged by educational value, distributional effect, recurrent affordability and resilience. Digital transformation will be credible when technology expands the capacity of teachers and institutions to support learning, rather than transferring risk and cost to households or using adoption as a proxy for success.

11. Limitations

As a critical narrative review, this article is intrinsically vulnerable to selection and interpretive bias despite the use of transparent searching, explicit eligibility criteria and critical appraisal. The search relied on live, web-accessible scholarly records and did not include complete subscription-database exports; relevant studies not discoverable through accessible indexes may therefore have been missed. English-language literature was prioritised, which may under-represent Bangla-language research and local evaluation reports. The evidence base is heterogeneous in design, population, technology and outcome, preventing quantitative synthesis and limiting direct comparison.

A large proportion of Bangladesh-specific studies were produced during or soon after the COVID-19 disruption, when online learning reflected emergency conditions rather than mature programme design. Cross-sectional surveys, convenience samples, self-reported outcomes and online data collection are common, creating risks of selection bias and over-representation of connected university students. Primary, secondary, madrasa, TVET, disability-inclusive and rural evidence is comparatively limited, while long-term learning outcomes, cost-effectiveness and implementation fidelity are rarely reported. The field is also changing quickly, particularly in relation to artificial intelligence, cybersecurity and blended-learning policy, so later evidence may alter the balance of conclusions.

Declaration of AI Use

This manuscript was prepared through the combined contributions of all author(s), including contributions to the study design, data, content development, results, interpretation, and related scholarly work. The author(s)

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Competing Interests

Authors have declared that they have no known competing financial interests or non-financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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